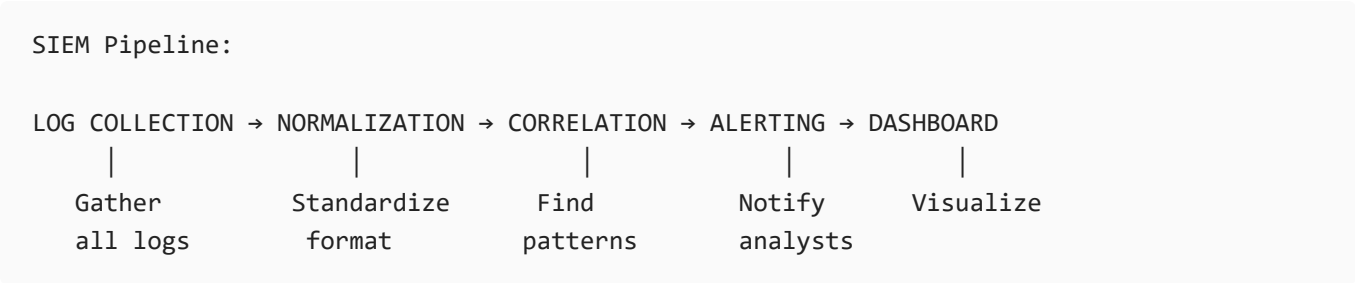


# 05 - SIEM

## 1: SIEM Fundamentals

**SIEM** = Security Information and Event Management

SIEM collects logs from multiple sources, normalizes them into a standard format, correlates events to detect threats, and alerts analysts. It's the central nervous system of a SOC.



### Key SIEM Concepts:

### SIEM / Log Analysis Basic Concepts

| Concept     | Meaning                    | Example                      |
|-------------|----------------------------|------------------------------|
| Log Source  | Where logs come from       | Windows, Firewall, Linux     |
| Index       | Storage container for logs | "windows_logs", "firewall"   |
| Event       | Single log entry           | One failed login             |
| Field       | Data element in event      | EventID, Username, IP        |
| Query       | Search command             | Find all failed logins       |
| Alert       | Automated notification     | Trigger when >10 failures    |
| Correlation | Connect related events     | Failed logins → then success |

### Popular SIEMs Platforms & Query Languages

| SIEM               | Query Language | Common In                |
|--------------------|----------------|--------------------------|
| Splunk             | SPL            | Enterprise, MSSPs        |
| Microsoft Sentinel | KQL            | Cloud/Azure environments |
| Elastic SIEM       | Lucene / EQL   | Open-source environments |
| IBM QRadar         | AQL            | Large enterprises        |
| Chronicle (Google) | YARA-L         | Google Cloud             |

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## 2: Splunk SPL Basics

**SPL** = Search Processing Language

### Basic Query Structure:

```
index=<index_name> <search_terms>
| command1
| command2
| command3
```

### Essential SPL Commands:

#### 1. Basic Search

```
# Search all Windows events
index=windows

# Search for specific EventID
index=windows EventID=4625

# Search with multiple conditions
index=windows EventID=4625 src_ip="45.142.212.61"

# Search with wildcard
index=windows user=admin*

# Search with time range
index=windows EventID=4625 earliest=-24h latest=now
```

#### 2. stats count by

**Purpose:** Count events grouped by field

```
# Count failed logins by source IP
index=windows EventID=4625
| stats count by src_ip

# Output:
# src_ip      | count
# 45.142.212.61 | 523
# 192.168.1.50  | 12
# 10.0.0.5     | 3

# Count failed logins by username
```

```
index=windows EventID=4625
| stats count by user

# Count by multiple fields
index=windows EventID=4625
| stats count by src_ip, user
```

### 3. table

**Purpose:** Display specific fields only

```
# Show only these columns
index=windows EventID=4625
| table _time, src_ip, user, dest

# Output:
# _time          | src_ip          | user  | dest
# 2024-03-15 10:23:45 | 45.142.212.61 | admin | DC01
# 2024-03-15 10:23:46 | 45.142.212.61 | admin | DC01
```

### 4. sort

**Purpose:** Order results

```
# Sort by count (descending = highest first)
index=windows EventID=4625
| stats count by src_ip
| sort -count

# Sort ascending (lowest first)
index=windows EventID=4625
| stats count by src_ip
| sort +count

# Sort by time
index=windows EventID=4625
| sort -_time
| table _time, src_ip, user
```

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## 3: KQL Basics (Microsoft Sentinel)

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## 4: Detection Queries

## Brute Force Detection

### Splunk:

```
index=windows EventID=4625
| stats count by src_ip
| where count > 50
| sort -count
```

## Password Spraying Detection

### Splunk:

```
index=windows EventID=4625
| stats dc(user) as unique_users, count by src_ip
| where unique_users > 10
```

## Successful Login After Failures

### Splunk:

```
index=windows (EventID=4625 OR EventID=4624)
| stats count(eval(EventID=4625)) as failures, count(eval(EventID=4624)) as successes by src_ip
| where failures > 10 AND successes > 0
```

## Suspicious PowerShell Detection

### Splunk:

```
index=windows EventID=4104
| search ScriptBlockText="*DownloadString*" OR ScriptBlockText="*-enc*" OR ScriptBlockText="*IEX*"
| table _time, ComputerName, ScriptBlockText
```

## New Admin Account Created

### Splunk:

```
index=windows EventID=4720
| join user [search index=windows EventID=4732 group_name="Administrators"]
| table _time, user, src_user
```

## Log Clearing Detection

## Splunk:

```
index=windows EventID=1102  
| table _time, user, ComputerName
```